Feolu Kolawole

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EDUCATION

Stanford University **Expected Graduation Date: June 2028**
B.S. in Computer Science

- **Coursework:** Data Structures & Algorithms, Operating Systems, Concurrency, Linear Algebra, Multivariable Calculus, Machine Learning, Computer Vision, Reinforcement Learning, Statistics, Combinatorics
- **Activities/Societies**: Optiver FutureFocus (Quant), Stanford XR (VP), Stanford AI Club, Stanford Neurotech

PROFESSIONAL EXPERIENCE

Microsoft **Redmond, WA**
Machine Learning Researcher *Incoming Jun 2026*

- Researching machine learning and computer vision techniques for MRI imaging, advised by Dr. Olesya Melnichenko.

Stanford Artificial Intelligence Laboratory **Stanford, CA**
Machine Learning Researcher (PI - Fei-Fei Li & Ehsan Adeli) *Jan 2026 - Present*

- Engineered a video-language pipeline to automatically monitor ICU patient behavior from continuous footage, enabling real-time health assessment of **2,656** critically ill patients without manual intervention.

Lead Machine Learning Researcher (PI - Ron Fedkiw) *Mar 2025 - Dec 2025*

- Drove development of CNN models to generate realistic 3D hair reconstructions across demographics, boosting accuracy to 96% and enabling deployment on lightweight devices.
- Built a highly robust extraction algorithm with OpenCV and SAM segmentation, capable of handling poor-quality images and achieving perfect segmentation on 98% of images.

Stanford Human Perception Lab **Palo Alto, CA**
Machine Learning Researcher (PI - Khizer Khaderi) *Jan 2026 - Present*

- Built a single model that predicts user actions from unlabeled video across **9 distinct environments**, achieving **85.2%** accuracy and enabling zero-shot action understanding without labeled training data.

Software Engineer *Dec 2024 - Jun 2025*

- Engineered an AR application on Snap Spectacles in collaboration with the Snapchat Prototyping Team, enabling analysis of **92%** of the available field of vision.
- Created a process to transform video into accurate 3D environments by integrating SLAM and point-cloud segmentation models, thereby optimizing output by **44%**.

PROJECTS

MoSV: Mixture-of-Steering Vectors for LLM Hallucination Mitigation | *Python, PyTorch*

- Proposed a framework that reduces LLM hallucinations by dynamically selecting from multiple learned correction vectors per prompt, improving factual accuracy by **+2.4pp** where prior methods gained only +0.3pp.
- Demonstrated that the framework automatically discovers distinct types of hallucinations without manual labeling, evaluated across **10,615** items spanning **8 knowledge domains**.

Power Lever: GPU-Efficient LLM Inference Gateway (Winner, Stanford Hackathon) | *Python*

- Built an inference gateway that dynamically routes LLM prompts to right-sized GPU hardware across **4 tiers**, reducing energy consumption by **75%** on simple queries while reserving high-end GPUs for complex tasks.

DYNAMO: Reinforcement Learning Portfolio Manager | *Python, Stable-Baselines3*

- Trained a reinforcement learning agent to automatically manage a portfolio across **10 asset classes**, achieving **15.76%** annualized returns with a **1.44 Sharpe ratio**, outperforming traditional strategies by **38–54%**.

Eous: Embodied AR Robot Assistant (Winner, Stanford Hackathon) | *Python, C#, Unity*

- Built a hands-free AR system integrating AR glasses, a smartphone, and a Raspberry Pi robot, enabling gesture and voice control with live camera feedback — all processed on-device with no cloud dependencies.

VR Healthcare Training Simulation (Winner, MIT Hackathon) | *Swift , RealityKit*

- Created a VR healthcare simulation on the Apple Vision Pro, enabling CPR and first aid training and pioneering SharePlay integration for collaborative learning with iPhone users.
- Showcased to **The Venture Reality Fund**, where the concept was acquired and carried forward.

Supreme Court Case Prediction (1st Place, Stanford Class of 500) (View Paper) | *Python, Numpy*

- Built a Bayesian court case prediction framework that achieved **73% top-3 accuracy** across **11 possible Supreme Court case outcomes** by leveraging observable case attributes.

- Featured as a winner and future example project in Stanford's CS109 course (selected from **500 students**).

SynchroSound: Facial Expression Based Song Selection | *Swift, API, Javascript, React, Objective-C*

- Built an iOS app that deciphers facial expressions to recommend mood-matching songs, integrating SwiftUI, UIKit, and SwiftData with Google Cloud Vision and Spotify's Web API.
- Processed **500+ test images** across multiple emotions to validate recommendations, improving user-song mood alignment **accuracy by 82%**.

Heap Allocator – Custom Computer Memory Management System | *C, x86 Assembly, GDB*

- Developed an algorithm that optimizes computer memory utilization (i.e. freeing up storage by **20%** across **100,000** requests).
- Implemented custom malloc, realloc, and free functions in C for heap-level memory management.

Shell - Custom Unix Shell Implementation | *C, C++, GDB, x86 Assembly*

- Built a command-line Unix Shell (computer operating system) that allows key tasks like file manipulation and redirection.
- Enhanced the Unix Shell to run **10+ programs** at once while testing across **100+ real-world** scenarios.

Medical MRI Image Reconstruction (View Paper) | *Python, Pytorch, Deep Learning*

- Engineered AI-based models for medical MRI image reconstruction with PyTorch, **cutting required scan times by 55–70%** while preserving quality images, and co-authoring a research paper on the work.
- Enhanced the MRI image quality beyond baseline deep learning models, **reducing reconstruction error by ~60%** and significantly improving detail preservation.

AgenTeX: Image to LaTeX (Winner, AgentOps Hackathon) | *Hugging Face, Python*

- Engineered an AI agent with OpenAI's Agents SDK to convert handwritten math to LaTeX with 91% accuracy.
- Featured as an AgentOps hackathon winner, with board member recommendation for commercial launch.

Music Recommendation System (View Paper) | *Python, SVD, Dimensionality Reduction, PCA*

- Designed a music recommendation system using SVD and PCA to analyze over **60 audio features** from **7,000 Spotify songs**, uncovering patterns in sound beyond traditional metadata.
- Implemented feature projection to identify similarities across **35 combinations of musical features**.

TECHNICAL SKILLS

Programming Languages: Python, Java, Javascript, Typescript, C, C#, C++, Swift, Assembly, HTML/CSS, R
Libraries/Frameworks: Next.js, React.js, Flask, FastAPI, Node.js, Tailwind CSS, Node.js, OpenGL, Git, Linux
Software: Github, Docker, Vim, Emacs, Pytorch, Visual Studio Code, XCode, Blender, Unity, Lens Studio, RStudio